

Heat Transfer

Conduction

Conduction is the process of heat transfer via a body/multiple bodies by collision of atoms. A fundamental equation that describes conduction is Fourier's Law of Conduction. In a bar of thickness A , and any point along the bar can be described as x units away from an origin, the equation states that

$$Q \propto A \frac{dT}{dx}$$
$$Q = -kA \frac{dT}{dx}$$

This equation means that for a change in temperature (say an increase in temperature) across a small distance dx , heat $[Q]$ flows negatively against this change in temperature, i.e. if going rightward across a bar has an increase in temperature, heat flows against this (to achieve thermal equilibrium). k is also the thermal conductivity of the material, in W/mK

Now, consider a slab of area A , and thickness X :

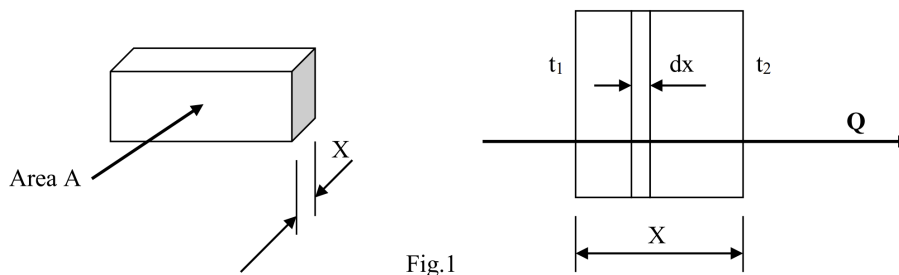


Fig.1

Say we heat it across the thickness X by apply two different temperatures, T_1 and T_2 , at opposite sides of the slab. Then, we can first rewrite Fourier's law in differential form:

$$Q dx = -kA dT$$

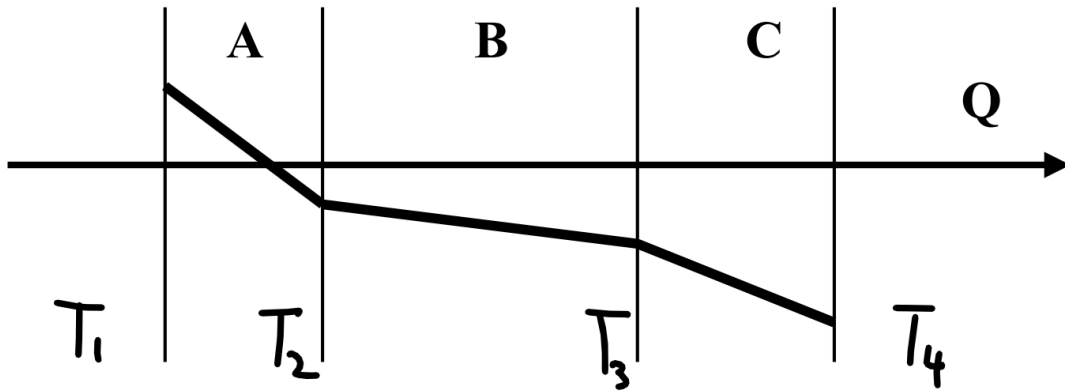
and solve for Q as a function of x with separation of variables:

$$\begin{aligned}
\int_0^X Q dx &= -kA \int_{T_1}^{T_2} dT \\
[Qx]_0^X &= -kA[t]_{T_1}^{T_2} \\
QX &= -kA(T_2 - T_1) \\
Q &= \frac{-kA(T_2 - T_1)}{X} \\
Q &= \frac{kA(T_1 - T_2)}{X}
\end{aligned}$$

which can be written as

$$Q = \frac{T_1 - T_2}{\frac{X}{kA}} \Rightarrow Q = \frac{\Delta T}{\frac{X}{kA}}$$

If we had multiple different material slabs next to one another, and we apply a temperature gradient again of T_1 to T_4 , like so,



We can write the heat transferred across each slab as

$$Q_A = \frac{k_A A (T_1 - T_2)}{X_A}, \quad Q_B = \frac{k_B A (T_2 - T_3)}{X_B}, \quad Q_C = \frac{k_C A (T_3 - T_4)}{X_C}$$

which can be written as

$$T_1 - T_2 = \frac{Q_A X_A}{k_A A}, \quad T_2 - T_3 = \frac{Q_B X_B}{k_B A}, \quad T_3 - T_4 = \frac{Q_C X_C}{k_C A}$$

but since the heat transferred through A must also be transferred to B and then C,

$$Q = Q_A = Q_B = Q_C$$

thus,

$$T_1 - T_2 = \frac{QX_A}{k_A A}, \quad T_2 - T_3 = \frac{QX_B}{k_B A}, \quad T_3 - T_4 = \frac{QX_C}{k_C A}$$

If we add these equations together, we get

$$\begin{aligned} T_1 - T_2 + T_2 - T_3 + T_3 - T_4 &= \frac{QX_A}{k_A A} + \frac{QX_B}{k_B A} + \frac{QX_C}{k_C A} \\ \cancel{T_1 - T_2} + \cancel{T_2 - T_3} + T_3 - T_4 &= Q \left(\frac{X_A}{k_A A} + \frac{X_B}{k_B A} + \frac{X_C}{k_C A} \right) \\ T_1 - T_4 &= Q \sum \frac{X_i}{k_i A} \end{aligned}$$

Thus, the heat transferred through a composite slab is

$$Q = \frac{T_1 - T_4}{\sum \frac{X_i}{k_i A}} = \frac{\Delta T}{\sum \frac{X_i}{k_i A}}$$

When there is a difference of temperatures, notice how heat does not immediately flow from the hotter end to the cooler end, but there is some resistance to it. We can think of this resistance being analogous to electricity, where we know from Ohm's Law that

$$I = \frac{V}{R} \Rightarrow \text{current} = \frac{\text{potential difference}}{\text{resistance}}$$

Likewise, across a slab,

$$\text{heat flow} = \frac{\text{temperature difference}}{\text{resistance}}$$

We already know a formula that relates heat flow and temperature difference,

$$Q = \frac{T_1 - T_2}{\frac{X}{kA}}$$

thus,

$$\text{thermal resistance (of a slab), } R_s = \frac{X}{kA}$$

Hence, the earlier equation for the heat transferred through a composite slab can be written as

$$Q = \frac{T_1 - T_4}{\sum R_s} = \frac{\Delta T}{\sum R_s}$$

In general, the formula can be written as $Q = \frac{\Delta T}{\sum R}$.

So far, we have only covered cuboidal slabs. If we had a hollow cylinder of some thickness heated from the inside out radially, or otherwise, the equation is

$$Q = \frac{T_1 - T_2}{\frac{1}{2\pi kL} \ln\left(\frac{r_2}{r_1}\right)} = \frac{\Delta T}{R_c}, \quad R_c = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi kL}$$

where it is assumed that $r_2 > r_1$ and $T_1 > T_2$, aka heat flows radially outward.

This equation is also the steady state heat transfer, and assumes no heat is trans

In a composite material cylinder, the formula is

$$Q = \frac{\Delta T}{\sum R_c}, \quad \sum R_c = \sum \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi kL}$$

So far, we can only find the heat transferred if the temperature at each wall is known.

However, it is more common that we only know the bulk temperature of the fluids separated by the slabs, and not the wall temperature themselves.

The equation for the heat transferred from a fluid to the wall is given as

$$Q = hA(T_f - T_w) = hA\Delta T$$

where h is the surface heat transfer coefficient, or film coefficient, in units W/m^2K .

In this case, the thermal resistance the boundary layer (between the fluid and wall) is $\frac{1}{hA}$.

Now, if we use the case where there are two fluids separated by a monolithic slab, we can treat each fluid as its own "slab", and use the formula for a composite slab, as

$$Q = \frac{\Delta T}{\sum R}$$

$$Q = \frac{T_i - T_o}{R_i + R_s + R_o}$$

$$Q = \frac{T_i - T_o}{\frac{1}{h_i A} + \frac{X}{kA} + \frac{1}{h_o A}}$$

Similarly, a composite slab separating two fluids would have the heat transferred be

$$Q = \frac{T_i - T_o}{\frac{1}{h_i A} + \sum \frac{X_j}{k_j A} + \frac{1}{h_o A}}$$

We can then rewrite the heat transferred through a composite slab separating two fluids in a similar form as $Q = hA\Delta T$,

$$Q = \frac{T_i - T_o}{\frac{1}{A} \left(\frac{1}{h_i} + \sum \frac{X_j}{k_j} + \frac{1}{h_o} \right)}$$

$$Q = A \times \frac{T_i - T_o}{\frac{1}{h_i} + \sum \frac{X_j}{k_j} + \frac{1}{h_o}}$$

$$Q = \frac{1}{\frac{1}{h_i} + \sum \frac{X_j}{k_j} + \frac{1}{h_o}} A (T_i - T_o)$$

Thus, the equation can be written as

$$Q = UA\Delta T$$

where

$$U = \frac{1}{\frac{1}{h_i} + \sum \frac{X_j}{k_j} + \frac{1}{h_o}}$$

$$\frac{1}{U} = \frac{1}{h_i} + \sum \frac{X_j}{k_j} + \frac{1}{h_o}$$

U is known as the overall heat transfer coefficient, given as so. Similarly, a monolithic slab would have the coefficient be

$$\frac{1}{U} = \frac{1}{h_i} + \frac{X}{k} + \frac{1}{h_o}$$

More generally, we can find $\sum R$ from $Q = UA\Delta T$, where

$$\frac{1}{UA} = \sum R$$

Thus, for a cylinder heated from inside to out by two fluids,

$$\frac{1}{UA} = \frac{1}{h_i A_i} + \sum \frac{\ln\left(\frac{r_o}{r_i}\right)}{2\pi k L} + \frac{1}{h_o A_o}$$

where $A_i = 2\pi r_i L$ and $A_o = 2\pi r_o L$.

To use the equation in $Q = UA\Delta T$, we pick either $Q = U_i A_i \Delta T$ or $Q = U_o A_o \Delta T$, as U may be based on either area.

If the cylinder is assumed to be thin walled, then $r_o \approx r_i$, and hence $A_o \approx A_i$. Thus, $\ln\left(\frac{r_o}{r_i}\right) \approx 0$, and

$$\frac{1}{U\cancel{A}} = \frac{1}{\cancel{h_i A}} + \frac{1}{\cancel{h_o A}}$$

$$\frac{1}{U} = \frac{1}{h_i} + \frac{1}{h_o}$$

During conduction, there may be some oil, scale or soot on the heating surfaces, which can cause additional heating resistance, known as the fouling resistance, R_f . For a unit area, this is known as the fouling factor, f .

Since a higher area causes a lower heating resistance, we can say that $R_f \propto \frac{1}{A}$, or that

$$R_f = \frac{f}{A}$$

$$f = R_f A$$

Hence, the units for R_f is K/W , and the fouling factor has units Km^2/W .

Thus, if we had a thin cylinder with some non-negligible debris on the walls, the heating coefficient is given as

$$\frac{1}{U} = \frac{1}{h_i} + \frac{1}{h_o} + f_i + f_o$$

(h has units W/m^2K)

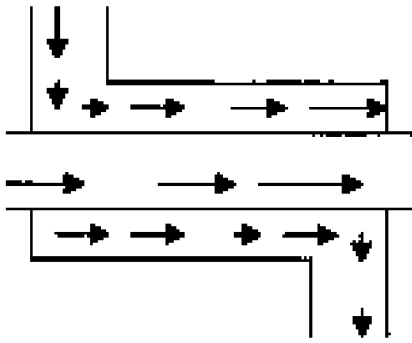
Practically, insulating materials are applied to slabs and tubing depending on the application. The process of applying insulating materials is known as lagging, and is used for the calculation of composite slabs and tubes.

(PROPERTIES OF LAGGING TESTED FOR MST)

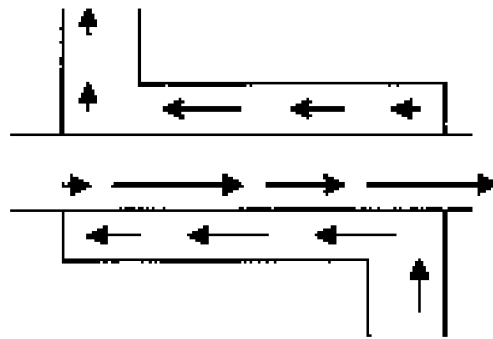
Convection & Heat Exchangers

Heat exchangers are devices that have two fluids, one hot and one cold, so as to heat the cold fluid or cool the hot fluid, through principles of convection. Heat exchangers can be set up in different ways to arrange the flow of the fluids.

In simple cases, fluids can either flow parallel or counter each other:

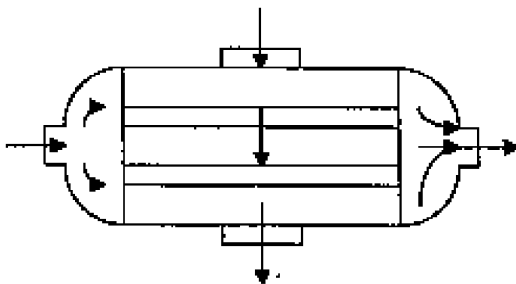


Parallel Flow

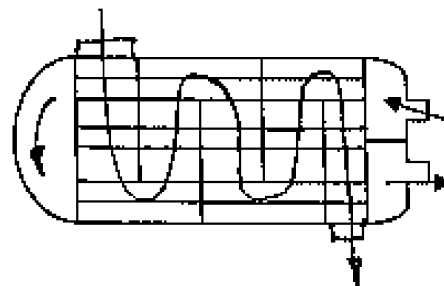


Counter Flow

Fluids may also flow perpendicularly to one another, either in cross or mixed flow, or a combination of them.

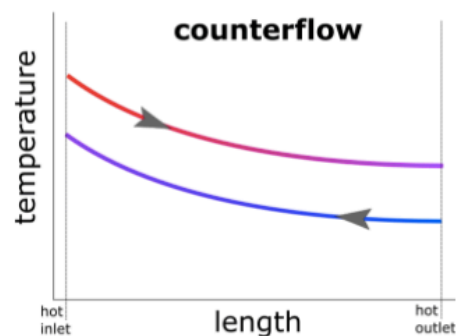
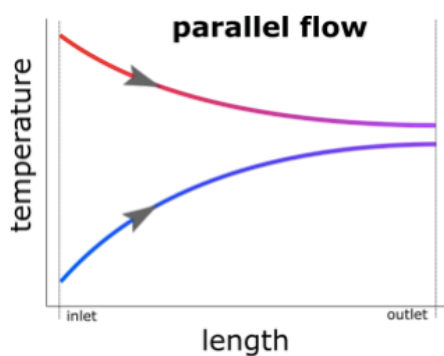


Cross Flow



Mixed Flow

From parallel and counter flow heat exchangers, we can measure the temperature difference across the length.



For parallel flow, the hot fluid starts out as the hottest possible, at the same side at where the cold fluid starts out at the coldest temperature. Along the length, both

temperatures tend towards thermal equilibrium, causing ΔT at each point along the length to drop.

For counter flow, the hottest and coldest temperature of the respective fluids are at opposite ends of the length, thus the hottest temperature will interact with the less cold fluid and the coldest temperature will interact with the less hot fluid. This causes ΔT to be roughly constant across the length of the exchanger.

[types of heat exchangers]

Generally, the heat transferred in a heat exchanger can be found as

$$Q = UA\Delta T_m = UA \times \frac{T_1 - T_2}{\ln\left(\frac{T_1}{T_2}\right)}$$

ΔT_m is known as the logarithmic mean temperature difference, or LMTD, representing the effective temperature difference of heat exchangers in parallel or counter flow.

T_1 and T_2 represent the difference in temperatures at each end of the length, the inlet and outlet respectively, which can be determined by the type of flow.

In cross of mixed flow heat exchangers, a correction factor is introduced, as

$$\text{Mean Temperature Difference} = \Delta T_m \times f$$

where f is a correction factor.

In the case where one of the fluids undergoes an isothermal process, the LMTD is the same for either flow. However, in most cases, LMTD is higher for counter flow than for parallel flow.

[advantages and other bs]

Fluids also transfer heat through convection. Recall the equation for heat transferred from a fluid to a wall:

$$Q = hA(T_f - T_w)$$

h , the heat transfer coefficient, is a function of the fluid and external properties.

More specifically, it is defined by the equation:

$$\frac{hd}{k} = F \left(\frac{\rho cd}{\mu} \right)^a \left(\frac{C_p \mu}{k} \right)^b$$

The equation equates one fluid quantity with two fluid quantities, where

$$\frac{hd}{k} = \text{Nu (Nusslet's Number)}$$

(a dimensionless parameter representing the ratio of convective to conductive heat transfer),

$$\frac{\rho c d}{\mu} = Re \text{ (Reynold's Number)}$$

(a parameter representing the fluid flow, where c is the velocity),

$$\frac{C_p [\text{specific heat capacity}] \mu}{k} = Pr \text{ (Prandtl's Number)}$$

(a parameter representing how heat diffuses from the fluid).

a and b are exponents that depend on the setup.

For non-circular pipes, an effective diameter value can be found as

$$d = \frac{4S [\text{flow area}]}{P [\text{wetted perimeter}]}$$

Typically, for turbulent flow, or more specifically non-laminar flow, one relationship to find Nusset's number is

$$Nu = 0.023 Re^{0.8} Pr^{0.4} [Re > 2000]$$

For laminar flow, Nu is usually constant, at

$$Nu = 3.65$$

For Nu, Re or Pr, we typically evaluate them at some mean temperature.

In most cases, where the fluid temp is close to the surface temperature, we use the mean temperature as the bulk temperature. In other words, the temperature we find these properties at is

$$T_b = \frac{T_i + T_o}{2}$$

If the difference between the fluid and surface temperatures are non-negligible, we use the mean film temperature, being

$$T_f = \frac{T_b + T_w}{2}$$

[types of heat exchangers]

Radiation

Thermal radiation is the process of heat transfer due to electromagnetic radiation. Thus, no medium is needed for radiation to occur.

When radiation strikes a body, it can either reflect, absorb or transmit (allow it to pass through) the radiation. The proportion of how the radiation is used afterwards can be represented by

$$\alpha + \rho + \tau = 1$$

where α , ρ and τ are the absorptivity, reflectivity and transmissivity fractions of the material. However, since most materials in engineering applications are not transmissive, τ can be assumed to be 0. Thus,

$$\alpha + \rho = 1$$

A blackbody is an object whose material completely absorbs any radiation. In other words, $\alpha = 1$. However, this only applies to ideal blackbodies. Although no ideal blackbodies exist, we can approximate some surfaces to it. For example, when a hot body is placed in a large surrounding, the surroundings can be approximated as a blackbody, like a steam pipe in a room. Another case is when a small opening leads to a larger opening, which makes the smaller area an approximate blackbody.

When a material absorbs radiation (or is generally hotter than the surroundings), it will radiate energy outward. The energy radiated is called the emissive power, E , per unit time per unit area, in units W/m^2 .

To compare an object to a blackbody, we use the emissivity, which is

$$e = \frac{E}{E_B}$$

where E_B is the emissive power of a blackbody of the same temperature.

Typically, rougher surfaces are able to absorb radiation better than smoother surfaces. They also have higher emissivity of radiation than smooth surfaces.

In calculations, we assume surfaces to have a constant emissivity and absorptivity over all wavelengths and temperatures. Objects with these assumptions are called grey bodies.

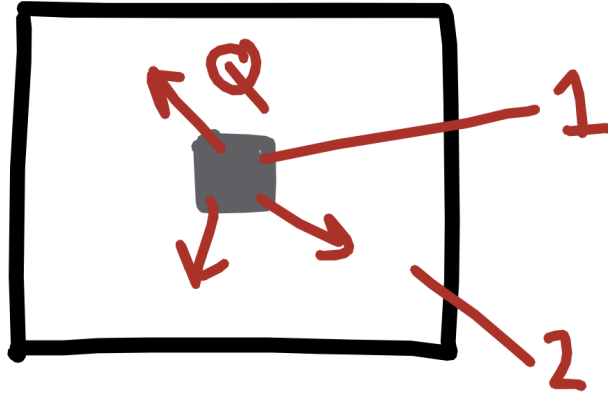
In contrast to a grey body, a surface which varies emissivity with wavelength and temperature is called a selective emitter. Selective emitters also selectively absorb radiation too, being selective absorbers.

To find emissivity, we can find the emissive power from the temperature of the blackbody, governed by the Stefan-Boltzmann Law.

$$E_B = \sigma T^4$$

where σ is the Stefan-Boltzmann constant, of $5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$. The temperature used in the formula is absolute, i.e. in kelvin.

Now, consider an example of a grey body transferring heat to a blackbody surroundings. Thus, $T_1 > T_2$.



The grey body emits the energy

$$\begin{aligned} E_1 &= e_1 E_B \\ E_1 &= e_1 \sigma T_1^4 \\ \text{Energy emitted}_1 &= e_1 A_1 \sigma T_1^4 \end{aligned}$$

while the blackbody emits

$$\begin{aligned} E_2 &= E_B \\ E_2 &= \sigma T_2^4 \\ \text{Energy emitted}_2 &= A_2 \sigma T_2^4 \end{aligned}$$

Firstly, we can assume that even though the blackbody emits $A_2 \sigma T_2^4$ of energy, only the grey body's area will absorb that, by the absorptivity fraction of the material too. In other words, the energy absorbed by the grey body from the blackbody is $\alpha_1 A_1 \sigma T_2^4$. Since it is a grey body, we assume that $e_1 = \alpha_1$, thus the energy absorbed by the grey body is $e_1 A_1 \sigma T_2^4$.

Thus, the heat transferred is

$$\begin{aligned} Q_{1 \rightarrow 2} &= \text{Energy emitted} - \text{Energy absorbed} \\ &= e_1 A_1 \sigma T_1^4 - e_1 A_1 \sigma T_2^4 \\ &= e_1 A_1 \sigma (T_1^4 - T_2^4) \end{aligned}$$

Notice how when $T_1 = T_2$, the heat transferred is 0, due to thermal equilibrium.

We can extend the heat transferred to be more general, by considering two grey bodies, which will give the equation

$$Q_{1 \rightarrow 2} = f_{1 \rightarrow 2} F_{1 \rightarrow 2} A_1 \sigma (T_1^4 - T_2^4)$$

where $f_{1 \rightarrow 2}$ is the interchange factor, and $F_{1 \rightarrow 2}$ is the geometric factor. Both are affected by the geometry of the setup.

The interchange factor is a measure of the effects of the emissivity of the bodies. In the case of blackbodies to blackbodies, the factor is 1.

The geometric factor is measure of how much radiation from body 1 falls on body 2.

Thus, if all of the radiation from 1 falls on 2, $F_{1 \rightarrow 2} = 1$.

Shown is a table of the factors given the setup.

Configuration	$f_{1 \rightarrow 2}$	$F_{1 \rightarrow 2}$
Infinite parallel planes	$\frac{1}{e_1^{-1} + e_2^{-1} - 1}$	1
1 enclosed by 2, 1 insignificant size	e_1	1
1 enclosed by 2, 1 significant size	$\frac{1}{e_1^{-1} + \frac{A_1}{A_2} (e_2^{-1} - 1)}$	1
Parallel discs x apart on common axis	$e_1 e_2$	$\frac{x^2 + r_1^2 + r_2^2 - \sqrt{(x^2 + r_1^2 + r_2^2)^2 - 4r_1^2 r_2^2}}{2r_1^2}$